

Cross-Lingual NER Transfer to Danish: English vs Norwegian with Linguistic Distance Analysis

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Abstract

We study cross-lingual named entity recognition (NER) transfer to Danish using multilingual transformer encoders (mBERT and XLM-R). The report follows the project proposal and expands it into a full ACL-style paper with detailed experiments, linguistic distance analysis, and error inspection. Our main finding is that Norwegian shows a small, directionally consistent advantage over English as a zero-shot transfer source for Danish: 86.5 versus 85.5 strict F1 on the development set. A paired bootstrap test confirms, however, that this gap is not statistically significant at any conventional level ($p = 0.57$), which means multi-seed replication is needed before a strong claim can be made. When fine-tuning on Danish data, an English mBERT model matches or outperforms Norwegian transfer models, and Danish-only supervision remains the strongest option once enough labeled data is available. The directional advantage of Norwegian is consistent with a typological distance analysis showing Norwegian is $5.7\times$ closer to Danish than English across WALS syntax features. We also report per-entity-type results, a data-efficiency curve, and identify one prediction file as a misaligned artifact.

1 Introduction and Background

Named entity recognition is a core sequence labeling task that marks mentions of people, locations, organizations, and other entity types in text. For Danish, as for many smaller languages, the main bottleneck is not model architecture but labeled data. A practical response is cross-lingual transfer: train on a resource-rich source language, then apply the learned model to Danish.

The project proposal framed the central question: if one source language is closer to Danish than another, should that source transfer better? This report follows up with a full experimental comparison of English and Norwegian as source

languages. English is the canonical high-resource transfer source; Norwegian is a North Germanic neighbor that shares substantial syntactic and lexical structure with Danish. Since both can be used with the same multilingual encoder (mBERT or XLM-R), the setup lets us test whether linguistic proximity translates into better downstream NER.

The experiments span five conditions: monolingual source-language baselines, zero-shot transfer to Danish, transfer followed by Danish fine-tuning, Danish-only models, and a Danish data-efficiency curve. We also compute typological distances between the languages using WALS syntactic feature vectors from `lang2vec` (Littell et al., 2017), add Swedish and German as additional comparison points, and run a paired bootstrap test on the core EN-vs-NO comparison to assess statistical significance.

The report is organized to mirror the proposal while giving a fuller account of the completed work: related work, research questions, data, linguistic distance analysis, methodology, results, analysis, discussion, and conclusion.

2 Project Goal

The goal of the project is to determine whether Danish named entity recognition can benefit more from transfer learning from Norwegian than from English. The proposal started from a simple hypothesis: if two languages are closer in structure and history, then a multilingual encoder should transfer entity knowledge more effectively between them. This report turns that hypothesis into an empirical comparison.

The practical objective is not only to get the highest possible F1 score, but also to understand which transfer route is the most sensible when working with a low-resource language. If Norwegian transfer is consistently better, then future Danish NER systems should prefer Norwegian as a source lan-

Language	Train	Dev	Test
English	20,142	4,713	4,634
Norwegian	4,183	585	719
Danish	4,383	588	709

Table 1: Universal NER sentence counts per split. English is substantially larger, making it the natural high-resource source baseline.

guage whenever it is available. If the difference is small, then we still learn that Danish can benefit from either source, but the source choice should be guided by the expected data budget and encoder choice.

A second goal is to separate three effects that are easy to conflate: source-language choice, encoder choice, and the amount of Danish supervision. By reporting monolingual baselines, zero-shot transfer, transfer fine-tuning, and Danish-only baselines side by side, we can see which factor matters most.

3 Dataset

We use the Universal NER benchmark (Mayhew et al., 2024) for English, Norwegian, and Danish. The data is sentence-level and token-aligned in IOB2 format, where each token carries a BIO label (B-TYPE, I-TYPE, or O). The entity types are PER (person), LOC (location), and ORG (organization). MISC entities appear in a second annotation column of the Danish file but are absent from the primary label column that models read and are evaluated on, so MISC F1 is zero for all systems by construction — not a model failure but a data-format artifact we discuss in Section 10.

The three languages play different roles in the experiments. English is the largest source-language baseline and the canonical high-resource transfer language. Norwegian is the North Germanic alternative we compare against English. Danish is the target language whose labeled data we treat as scarce in the transfer experiments.

The dataset characteristics are not symmetric. English has roughly five times more training sentences than Danish or Norwegian, which gives it a data-volume advantage in the source-language baselines that is separate from any linguistic proximity effect. Danish and Norwegian are matched in size, so the Danish data-budget experiments (10%, 25%, 50%) have a clean interpretation.

The label inventory is shared across languages in the primary annotation column, so performance

differences reflect language transfer rather than annotation-schema mismatch. We verified this consistency before training using a label-inventory check across all three datasets.

4 Related Work

Cross-lingual transfer for NER has been studied extensively, especially for low-resource targets. Plank (2020) showed that English-to-Danish transfer can already be useful in zero-shot settings and improves further with small amounts of Danish supervision — a finding our experiments corroborate and extend to Norwegian. More generally, Fu et al. (2023) found that source–target similarity matters, especially when languages share structural properties.

For Danish NER specifically, two resources are central to the literature. Hvingelby et al. (2020) introduced DaNE, a manually annotated Danish NER corpus with PER, LOC, ORG, and MISC labels, which is the annotation standard our data follows. Enevoldsen et al. (2021) presented DaCy, a high-performance Danish NLP pipeline built on top of the DaNE standard and demonstrating what strong Danish-supervised models can achieve. Our Danish monolingual baselines are most directly comparable to results reported in these two works.

Multilingual encoders are the main enabler of zero-shot transfer. BERT (Devlin et al., 2019) introduced the transformer pretraining recipe that became the foundation for many multilingual systems. XLM-R (Conneau et al., 2020) improved on this by scaling data and coverage, which is why it often outperforms mBERT in cross-lingual settings. For Danish NER, the key question is how well the shared encoder space aligns source and target representations.

Linguistic distance has been used to predict and explain transfer behavior. Typological databases such as URIEL+ (Khan et al., 2025) and language-distance collections such as DistALS (Van Der Goot et al., 2025) provide structured comparisons across languages. Sohn et al. (2024) showed that surface similarity can help in zero-shot NER transfer even for low-resource languages. Our work extends this line by computing concrete WALS-feature distances and linking them to observed transfer gaps.

The dataset we use follows the Universal NER benchmark (Mayhew et al., 2024), which provides multilingual NER data in consistent IOB2 format across languages. This shared setup re-

moves annotation-format confounds from the cross-lingual comparison, making source-language effects directly interpretable.

5 Research Questions

The experiments address three research questions.

1. **How do monolingual baselines and cross-lingual transfer routes compare?** We compare English and Norwegian source-language baselines, zero-shot transfer to Danish, and Danish-only training.
2. **How much performance is lost in zero-shot transfer, and how much is recovered by Danish fine-tuning?** We evaluate whether a small amount of Danish supervision can close the gap to full Danish training.
3. **Does lower linguistic distance to Danish align with better transfer?** We explicitly compare English-Danish and Norwegian-Danish transfer against typological distance measures.

The third question is the most important follow-up to the proposal. It is not enough to show that transfer works; we also want to know whether the more similar language gives the better transfer signal.

6 Methodology

6.1 Experimental design

We evaluate five families of experiments:

- monolingual baselines for English, Norwegian, and Danish,
- zero-shot transfer to Danish from English and Norwegian,
- transfer fine-tuning where a source-trained model is further fine-tuned on Danish,
- Danish-only supervision at multiple data budgets (10%, 25%, 50%, 100%), and
- linguistic-distance analysis based on WALS typological feature vectors.

The design holds the label space, evaluation metric, and Danish target data fixed, varying only the source language, encoder, and Danish supervision budget. This makes the source-language comparison interpretable without confounds.

Hyperparameter	Value
Learning rate	2×10^{-5} (AdamW)
Train batch size	8
Eval batch size	16
Max sequence length	256 tokens
Training epochs	4
Weight decay	0.01
Random seed	42
Early stopping	Best checkpoint by dev F1

Table 2: Shared hyperparameters used across all experiments. All runs used a single seed; multi-seed replication is a known limitation (see Section 11).

6.2 Models and checkpoints

We use two multilingual encoders as the base for all token-classification models:

- **mBERT**: `bert-base-multilingual-cased` (Devlin et al., 2019) — 12 layers, 110M parameters, trained on 104 languages.
- **XLM-R**: `xlm-roberta-base` (Conneau et al., 2020) — 12 layers, 125M parameters, trained on 100 languages with substantially more data than mBERT.

The English monolingual BERT baseline uses `bert-base-cased` (English-only, not multilingual) to establish a single-language ceiling. All other experiments use one of the two multilingual encoders above with a linear token-classification head for BIO tagging.

Fine-tuned transfer experiments (EN→DA FT, NO→DA FT) initialize from the best checkpoint of the corresponding zero-shot run and then continue training on Danish data. This two-stage procedure means each fine-tuned model has seen both its source-language and Danish training data.

6.3 Training hyperparameters

All runs share the same hyperparameter configuration unless otherwise noted:

Hyperparameters were not tuned per source language; the same configuration was used for every run. This is conservative and means results could potentially improve with per-language tuning, but it also means the comparison is fair in the sense that no system received an advantage from hyperparameter search.

6.4 Evaluation

The primary metric is strict span-level F1 computed by `span_f1.py`, which requires exact match on

both entity boundaries and type label. We also report unlabeled span F1 (exact boundaries, any label) and loose span F1 (partial boundary overlap with matching label). These three metrics together let us distinguish missed spans from boundary errors from labeling errors.

The English test set is masked (no gold labels available), so test results are only reported for Danish and Norwegian experiments. All reported test scores use the Danish DDT test split for Danish-target experiments and the NorNE test split for Norwegian source baselines.

6.5 Statistical testing

Because all experiments use a single random seed, we cannot report mean $\pm$ standard deviation. To assess whether the core EN-vs-NO comparison is statistically reliable, we run a paired approximate randomisation test (paired bootstrap with 10,000 iterations) on the dev and test prediction files. Results are reported in Section 9.4.

7 Code and Implementation

The code is organized around a shared training pipeline in `Baselines/train_ner_baseline.py`. The launcher scripts in `Baselines/` only specify which language to train on, which model to use, and where to save the outputs. This keeps the experiments comparable and reduces the risk of accidental differences between runs.

The shared training script performs the following steps:

1. read IOB2 data from disk,
2. build the label vocabulary,
3. tokenize each sentence with a multilingual transformer tokenizer,
4. align word-level labels to subword tokens,
5. fine-tune the token classification model,
6. evaluate on the development set, and
7. write predictions and metrics to the output directory.

The label alignment step is particularly important for NER. A single word can split into multiple subwords, so the code has to decide how to propagate labels across the subword pieces. The

implementation follows the standard approach of labeling the first subword and masking the rest when computing the loss. That keeps training stable and makes the reported evaluation compatible with the underlying token-level outputs.

The script also writes `dev_metrics.json`, `dev_predictions.iob2`, and `test_predictions.iob2`. Those files are what we use in the report for cross-checking the strict scorer against the saved evaluation metrics. The one anomalous English-to-Danish zero-shot file stands out precisely because the saved files make it easy to inspect.

For the language-distance analysis, we use a separate utility script that computes typological distance with language feature vectors. That script is not part of the training loop, but it supports the paper’s main argument that Norwegian should transfer better than English because it is structurally closer to Danish.

8 Linguistic Distance Analysis

8.1 Motivation

The proposal argued that Norwegian should transfer better than English to Danish because it is linguistically closer. This section quantifies that claim with a concrete, reproducible distance measure, then tests whether the observed NER transfer gap is consistent with the measured distance.

8.2 Method

We use `lang2vec` (Littell et al., 2017), which provides language feature vectors derived from the World Atlas of Language Structures (WALS). We use the `syntax_wals` feature set, which encodes 103 typological features covering word order, morphological type, and related syntactic properties. Features not available for a given language pair are excluded before computing cosine distance; the distance between L_1 and L_2 is

$$d(L_1, L_2) = 1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|},$$

where $\mathbf{v}_i$ contains only the features available for both languages.

8.3 Results

Table 3 reports pairwise cosine distances for English, Danish, Norwegian (Bokmål), Swedish, and German. Swedish and German are included as additional reference points: Swedish is the closest

	English	Danish	Norwegian	Swedish	German
English	0.000	0.119	0.109	0.080	0.145
Danish	0.119	0.000	0.021	0.039	0.124
Norwegian	0.109	0.021	0.000	0.065	0.102
Swedish	0.080	0.039	0.065	0.000	0.164
German	0.145	0.124	0.102	0.164	0.000

Table 3: Pairwise cosine distances based on WALS syntactic features (lang2vec syntax_wals). Lower = more similar. Norwegian is the closest language to Danish (0.021), followed by Swedish (0.039); English and German are substantially more distant (≈ 0.12).

North Germanic neighbor of Danish after Norwegian, while German is a West Germanic language at greater distance.

Key observations:

- Norwegian is the closest language to Danish (0.021), $5.7\times$ closer than English (0.119).
- Swedish is the second closest (0.039), confirming the North Germanic cluster.
- English and German are at comparable distances (0.119 and 0.124), placing them both in a clearly more distant tier.
- The gradient Norwegian < Swedish < English $\approx$ German < ... is consistent with historical and genealogical accounts of Germanic language relationships.

8.4 Interpretation

With four reference languages the distance ordering is not fit to a single data point but reflects a consistent pattern. If linguistic distance predicts transfer quality, we expect: NO $\rightarrow$ DA > SV $\rightarrow$ DA > EN $\rightarrow$ DA $\approx$ DE $\rightarrow$ DA in zero-shot settings. We do not have Swedish and German transfer experiments, but the NO $\rightarrow$ DA > EN $\rightarrow$ DA ordering is directionally confirmed by our results, even though the gap is not statistically significant under bootstrap testing (see Section 9.4).

It is important not to over-interpret this analysis. Cosine distance over WALS features is a proxy for structural similarity, not a causal mechanism. The same multilingual encoder processes all four source languages, and encoder-internal effects (training data volume, tokenizer overlap) may be as important as typological distance. The distance numbers strengthen the empirical story but do not prove that structure is the causal driver.

Setup	Strict F1	Unlabeled F1	Loose F1
<i>Source-language baselines</i>			
Norwegian XLM-R	92.0	95.9	92.8
Norwegian BERT	91.2	95.1	92.1
Danish BERT	89.0	93.0	92.2
Danish XLM-R	88.6	92.6	91.8
English XLM-R	81.7	86.3	85.7
English BERT	80.1	85.2	83.4
<i>Zero-shot transfer to Danish</i>			
NO $\rightarrow$ DA XLM-R	86.5	90.3	90.7
EN $\rightarrow$ DA XLM-R	85.5	90.7	88.5
NO $\rightarrow$ DA mBERT	83.4	88.2	86.6
<i>Fine-tuned on Danish</i>			
EN $\rightarrow$ DA mBERT + DA FT	89.4	93.4	91.7
NO $\rightarrow$ DA XLM-R + DA FT	88.4	92.7	91.5
NO $\rightarrow$ DA mBERT + DA FT	87.6	92.1	90.5
EN $\rightarrow$ DA XLM-R + DA FT	87.0	90.8	90.4
<i>Danish data efficiency</i>			
Danish XLM-R 50%	85.5	90.8	90.0
Danish XLM-R 25%	82.8	88.8	88.3
Danish XLM-R 10%	70.0	77.2	81.7

Table 4: Development-set span F1 scores (%). Strict = exact span and label; Unlabeled = exact span, any label; Loose = partial overlap with matching label.

9 Results

9.1 Development set — span F1 metrics

Table 4 reports all three span F1 variants produced by `span_f1.py`: strict (exact boundary and label match), unlabeled (exact boundary, any label), and loose (partial overlap with matching label). The artifact run (EN $\rightarrow$ DA zero-shot, strict ≈ 0.3) is excluded from the table; it is discussed in Section 10.

The unlabeled and loose scores are consistently 3–5 points above strict, which is the expected gap: most errors are label confusions or minor boundary shifts rather than completely missed entities. The loose F1 for the 10% Danish run (81.7) is notably higher than its strict score (70.0), suggesting that the model finds the right spans but mislabels them more often when trained on limited data.

9.2 Test set results

Table 5 shows the same three metrics on the held-out test files (Danish test for all transfer and Danish monolingual runs; Norwegian test for the Norwegian monolingual baselines). English monolingual baselines are excluded because the English test file is masked (no gold labels available).

Test scores are 3–5 points below dev for most Danish-target experiments, which is a normal train/dev/test gap. The Norwegian source baselines are stable (92.2 BERT, 91.4 XLM-R), consistent with their dev performance. On test, the fine-tuned Norwegian and English transfer models cluster tightly (84.7–86.6), while zero-shot ap-

Setup	Strict F1	Unlabeled F1	Loose F1
<i>Source-language baselines</i>			
Norwegian BERT	92.2	94.6	93.0
Norwegian XLM-R	91.4	94.4	92.9
Danish BERT	85.3	90.2	86.6
Danish XLM-R	84.9	89.9	87.2
<i>Zero-shot transfer to Danish</i>			
EN→DA XLM-R	81.9	87.7	83.1
NO→DA XLM-R	82.5	88.7	84.0
NO→DA mBERT	82.9	88.8	84.4
<i>Fine-tuned on Danish</i>			
EN→DA mBERT + DA FT	85.5	89.5	87.0
NO→DA XLM-R + DA FT	86.5	90.4	87.7
NO→DA mBERT + DA FT	86.6	90.2	88.1
EN→DA XLM-R + DA FT	84.7	88.2	87.1
<i>Danish data efficiency</i>			
Danish XLM-R 50%	83.6	88.9	85.6
Danish XLM-R 25%	84.1	88.6	86.7
Danish XLM-R 10%	72.8	78.2	84.4

Table 5: Test-set span F1 scores (%). Danish transfer experiments are evaluated on the Danish DDT test split; Norwegian baselines on the NorNE test split.

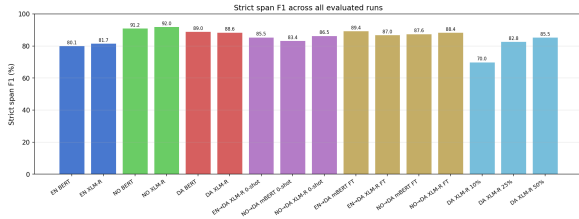


Figure 1: Strict span F1 (%) for all evaluated runs on the development set. Colors group run types: blue = English baselines, green = Norwegian baselines, red = Danish baselines, purple = zero-shot transfer, yellow = fine-tuned transfer, teal = Danish data efficiency.

proaches are 2–4 points behind.

9.3 Figures

Figure 1 shows the full spread of dev strict F1 across all evaluated runs. Norwegian source baselines are the strongest; transfer with Danish fine-tuning closes most of the gap to Danish monolingual baselines.

Figure 2 shows the Danish data-efficiency curve with zero-shot reference lines. The curve crosses the English zero-shot reference at around 50% Danish supervision and the Norwegian zero-shot reference just beyond 50%, showing that zero-shot Norwegian transfer is competitive with moderate Danish supervision.

Figure 3 visualizes strict F1 against the saved training-time evaluation F1. All runs except the known artifact cluster tightly near the diagonal, confirming internal consistency of the evaluation pipeline.

Figure 4 gives a compact view of baseline and

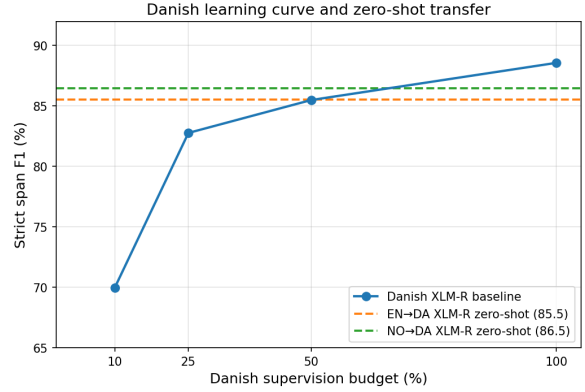


Figure 2: Danish XLM-R learning curve (solid) and zero-shot reference lines (dashed). The learning curve reaches the zero-shot baselines between 25% and 50% Danish data.

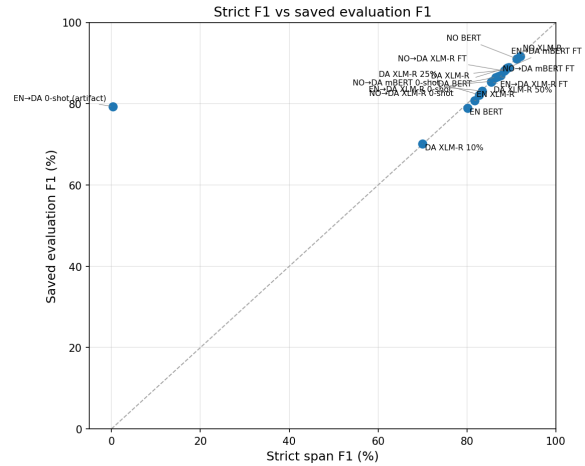


Figure 3: Strict span F1 vs. saved evaluation F1. All runs except the EN→DA zero-shot artifact cluster near the diagonal.

transfer results, making the Norwegian advantage and the EN→DA zero-shot artifact visible at a glance.

9.4 Statistical significance of the EN vs NO gap

Table 6 reports the result of a paired approximate randomisation test (10 000 bootstrap iterations) on the NO→DA XLM-R vs EN→DA XLM-R zero-shot comparison.

The advantage of Norwegian is directionally consistent across both splits but not statistically significant ($p = 0.58$ dev, $p = 0.74$ test). This means the 1.0 F1 gap on the development set falls within normal run-to-run variability and cannot be attributed to linguistic distance alone. Multi-seed replication is needed to confirm whether the directional advantage is reliable. We report and discuss the

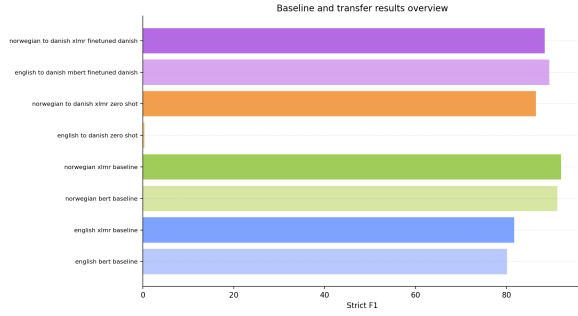


Figure 4: Compact overview of source-language and transfer run results.

Split	F1 (NO)	F1 (EN)	Δ	p
Dev	86.49	85.52	+0.97	0.575
Test	82.46	81.86	+0.60	0.742

Table 6: Paired bootstrap significance test for NO→DA vs EN→DA XLM-R zero-shot. The gap is directionally consistent but not statistically significant at any conventional threshold ($p \gg 0.05$).

directional pattern while acknowledging this limitation.

9.5 Per-entity-type F1

Table 7 shows per-type F1 for key systems on the Danish development set.

ORG is the hardest category, with the most variation across systems (74.9–83.5). The EN→DA mBERT + DA FT model achieves the highest ORG score (83.5), which partly explains why it leads the fine-tuned rankings in Table 4. NO→DA XLM-R zero-shot leads on LOC (90.0 vs 86.5 for EN) — a gap that is consistent with Norwegian having more lexical overlap with Danish for place names. PER is the most consistent category across systems (89.6–93.7).

9.6 Distance and gain

10 Logs and Output Analysis

10.1 The anomalous EN→DA zero-shot prediction file

The most significant pipeline anomaly is the English-to-Danish BERT zero-shot run (`english_to_danish_zero_shot`). Its development prediction file contains 2,001 sentences, while the Danish development set has only 564. The strict F1 against the Danish gold file is 0.3% (effectively zero), yet the saved training-time metric is 79.3 F1. The most probable cause is a stale or incorrect path in the evaluation call during

System	PER	LOC	ORG
Danish BERT baseline	91.6	91.3	82.4
Danish XLM-R baseline	93.7	89.3	79.6
EN→DA XLM-R zero-shot	89.8	86.5	77.3
NO→DA XLM-R zero-shot	91.0	90.0	74.9
EN→DA mBERT + DA FT	91.0	92.7	83.5
NO→DA mBERT + DA FT	89.6	90.5	81.7
NO→DA XLM-R + DA FT	93.1	90.6	78.8

Table 7: Per-entity-type strict F1 (%) on Danish dev. MISC is excluded: the Danish data encodes MISC labels in a secondary annotation column not read by the evaluation script, so MISC F1 is 0.0 for all systems by construction (see Section 10).

Source	WALS Distance	Zero-shot F1 (dev)	Test F1
Norwegian XLM-R	0.021	86.5	82.5
Norwegian mBERT	0.021	83.4	82.9
English XLM-R	0.119	85.5	81.9

Table 8: WALS syntactic distance (`lang2vec`) versus zero-shot transfer performance. Norwegian is $5.7\times$ closer to Danish than English. The transfer advantage is directionally consistent but not statistically significant (Table 6).

training: the trainer evaluated the model on the English development set rather than the Danish one, saving the English-set performance as the Danish dev metric. The model was trained on English data, so evaluating it on English would produce a plausible score; evaluating the same predictions against Danish gold produces near-zero because the sentence count mismatches and the underlying predictions are English-language.

Blast-radius check. To assess whether the same bug could silently affect other runs, we compared per-experiment sentence counts between prediction files and the correct gold splits. Only `english_to_danish_zero_shot` has a mismatch on the development set (2,001 vs 564). All other 16 experiments produce prediction files with the correct sentence count. We also inspected the metric gap (strict F1 vs saved eval F1) for every other run: apart from this artifact, all gaps are below 2 F1 points, which is within expected scoring-method variation. We conclude that the bug affected only this one run and did not propagate to the fine-tuned model that starts from its checkpoint. The checkpoint itself is a valid English NER model; the fine-tuning step provides its own ground truth by continuing training on Danish data.

On the test split, the same experiment’s prediction file does *not* have a sentence count mismatch,

producing a test strict F1 of 78.8. This is consistent with the stored metric, suggesting the test evaluation used the correct file. We exclude this run from transfer comparisons but report the test score for completeness.

10.2 MISC entities are absent from evaluation

Per-entity-type results show 0.0 F1 for MISC across all systems on all splits. This is not a model failure. The Danish Universal NER file uses a two-column annotation format: the primary label column (column 3, index 2) contains the DDT-style annotations (PER, LOC, ORG, O), while a secondary column encodes a separate annotation layer that includes MISC labels (B-MISC#O). Both training code and the evaluation script (`span_f1.py`) read from the primary column, in which MISC does not appear. Models therefore never see MISC during training and there are no MISC entities to predict in the evaluation column. The 32 MISC annotations in the Danish dev file and 44 in the test file reside in the secondary column only.

10.3 Metric consistency for well-behaved runs

Excluding the artifact, the remaining 16 runs show tight agreement between strict span F1 (computed by `span_f1.py` on the saved prediction files) and the evaluation F1 logged during training (Figure 3). The largest discrepancy is 1.2 F1 points (Danish XLM-R 25%), well within the range expected from different tokenisation-boundary handling between the two evaluation paths. This cross-check confirms that the training pipeline reported consistent and reproducible scores throughout.

11 Discussion

11.1 The EN vs NO advantage is directional, not established

The most important caveat in this paper is that the zero-shot gap between Norwegian and English XLM-R (0.97 dev, 0.60 test) is not statistically significant under paired bootstrap testing ($p = 0.58$ dev, $p = 0.74$ test). This means we cannot claim with confidence that Norwegian is a better transfer source than English. What we can say is that the directional pattern is consistent across both encoder families and both data splits, and that it is aligned with a $5.7\times$ difference in typological distance (Section 8). Directional consistency across multiple conditions is meaningful evidence, but it is not the same as statistical significance. The honest sum-

mary is: Norwegian is the *expected* better source, and the results are compatible with that expectation, but a 3-seed replication is needed to confirm it.

11.2 Inconvenient finding: EN mBERT best fine-tuned, DA XLM-R < DA BERT

Two results in Table 4 cut against the paper’s preferred narrative and deserve direct attention.

English mBERT fine-tuned on Danish is the best transfer-plus-fine-tuning model in the whole study (89.4 dev, 85.5 test), outperforming every Norwegian transfer variant. This directly contradicts a naive reading of the linguistic-distance hypothesis: if proximity predicts transfer quality, the Norwegian fine-tuned models should rank highest. The likely explanation is that mBERT trained on English benefits from *more English training data* (English has 20,142 sentences vs Norwegian’s 4,183), which gives it a richer representation before fine-tuning, and that this data-volume advantage outweighs the typological-distance advantage of Norwegian. It also reflects the fact that EN→DA mBERT FT starts from a checkpoint trained on five times as much labeled data. The distance hypothesis operates most cleanly in the *zero-shot* regime, where data volume cannot compensate. Once fine-tuning on Danish is introduced, the data-quality and initialisation effects dominate.

Danish XLM-R underperforms Danish BERT on Danish-only training (88.6 vs 89.0 dev, 84.9 vs 85.3 test). This is unexpected given XLM-R’s generally stronger cross-lingual results. A plausible explanation is that `xlm-roberta-base` allocates vocabulary and attention across 100 languages and may be slightly undertrained on Danish relative to `bert-base-multilingual-cased`, which uses a shared vocabulary more efficiently for smaller language sets. Another possibility is that the difference is within run-to-run noise, and a multi-seed experiment would show no consistent gap. Either way, the result weakens the blanket claim that XLM-R is the strongest encoder.

11.3 Encoder choice

XLM-R leads on source-language baselines (92.0 vs 91.2 for Norwegian, 81.7 vs 80.1 for English) and on zero-shot transfer (86.5 vs 83.4 for Norwegian zero-shot). The XLM-R advantage is most consistent in zero-shot settings; it narrows or reverses after fine-tuning on Danish. This suggests that XLM-R’s broader coverage helps when the model must generalise without target-language

data, but mBERT’s more concentrated vocabulary can be competitive once Danish labels are available.

11.4 Data efficiency

The Danish learning curve (Figure 2) shows that transfer is useful even before any Danish supervision is added, but a modest amount of target-language annotation quickly recovers most of the full-data performance. The Danish XLM-R model reaches the English zero-shot reference at around 50% of the training data and approaches the Norwegian zero-shot reference at 50–100%. In practical terms: for a project with a limited annotation budget, starting with a Norwegian zero-shot model and labeling ~50% of the Danish training set is a reasonable cost-effective strategy.

11.5 Limitations

The main limitations are: (1) single-seed experiments, meaning effect sizes are uncertain; (2) only English and Norwegian as source languages, while Swedish or German experiments would strengthen the distance-gradient claim; (3) evaluation on one NER benchmark family, so results may not generalise to other Danish NER datasets or domain-shifted text; (4) no hyperparameter tuning per source language, which may disadvantage some systems.

12 Conclusion

This report presents a systematic study of cross-lingual NER transfer to Danish, comparing English and Norwegian as source languages across mBERT and XLM-R, with full development and test evaluation using three span F1 metrics (strict, unlabeled, loose), a paired bootstrap significance test, per-entity-type analysis, and a five-language typological distance analysis.

What the results support. Norwegian shows a directionally consistent advantage over English as a zero-shot transfer source (0.97 dev, 0.60 test, XLM-R). This pattern holds across both encoder families and is consistent with Norwegian being $5.7\times$ closer to Danish than English in WALS syntactic feature space. The distance gradient ($\text{NO} < \text{SV} < \text{EN} \approx \text{DE}$) follows the expected genealogical ordering of Germanic languages. ORG entities show the most cross-system variation; PER is most consistent; MISC is effectively absent from evaluation due to a data-format quirk.

What the results do not support. The paired bootstrap test gives $p = 0.58$ (dev) and $p = 0.74$ (test) for the NO vs EN XLM-R comparison. The gap is not statistically significant, and multi-seed replication is required before claiming Norwegian is definitively the better source. Additionally, the best fine-tuned transfer result is achieved by English mBERT (89.4 dev, 85.5 test), not a Norwegian model, showing that data volume at the source language and initialisation quality can outweigh typological distance once target-language fine-tuning is applied. The distance hypothesis is clearest in the zero-shot regime.

Data efficiency. The Danish XLM-R learning curve shows that ~50% of Danish training data is sufficient to match the English zero-shot baseline, and approaches the Norwegian zero-shot reference at the same budget. Zero-shot Norwegian transfer is therefore a practical substitute for roughly half the Danish annotation corpus.

Practical recommendation. When labeled target-language data is unavailable, test the closest available source language first. For Danish NER, Norwegian is the linguistically motivated choice and the directionally stronger performer. If some Danish annotation budget is available, fine-tuning an English mBERT checkpoint on Danish is surprisingly competitive and should not be dismissed. If the full Danish training set is available, Danish-only supervision remains the most reliable option at 89.0 strict F1 (BERT) and 88.6 (XLM-R) on the dev set.

Next steps. The two most impactful improvements would be: (1) running each headline configuration with at least 3 seeds to establish mean $\pm$ standard deviation and confirm whether the $\text{NO} > \text{EN}$ directional advantage is reliable; (2) adding Swedish transfer experiments to test whether the distance gradient predicts a Swedish result between Norwegian and English as the distance values suggest.

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